

Q. 1 (Fun with distributions). We first consider a random variable $X \sim \text{Expon}(\lambda)$ and $Z = 1 - e^{-\lambda X}$.

(a) We have

$$\begin{aligned}\mathbb{E}(Z) &= 1 - \mathbb{E}(e^{-\lambda X}) \\ &= 1 - \int_0^\infty e^{-\lambda x} \cdot \lambda e^{-\lambda x} dx \\ &= 1 - \frac{1}{2} \int_0^\infty 2\lambda e^{-2\lambda x} dx \\ &= 1 - \frac{1}{2} = \boxed{\frac{1}{2}}\end{aligned}$$

where the third equality follows since $x \mapsto 2\lambda e^{-2\lambda x}$ ($x \geq 0$) is the density of the exponential distribution with parameter 2λ .

(b) Since $X \sim \text{Expon}(\lambda)$, we have $\mathbb{P}\{X \leq x\} = 1 - e^{-\lambda x}$ for $x \geq 0$. Hence,

$$\begin{aligned}\mathbb{P}\{Z \leq z\} &= \mathbb{P}\{1 - e^{-\lambda X} \leq z\} = \mathbb{P}\{1 - z \leq e^{-\lambda X}\} \\ &= \mathbb{P}\{X \leq -\frac{1}{\lambda} \log(1 - z)\} = 1 - e^{(-\lambda)(-\frac{1}{\lambda} \log(1 - z))} = z\end{aligned}$$

for $z \in [0, 1[$. (Note that $z \in [0, 1[$ if and only if $-\frac{1}{\lambda} \log(1 - z) \in [0, +\infty[$.) Hence,

$$\mathbb{P}\{Z \leq z\} = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{if } 0 \leq z < 1 \\ 1 & \text{if } z \geq 1 \end{cases}$$

Therefore, the density f of Z is given by

$$f(z) = \frac{d}{dz} \mathbb{P}\{Z \leq z\} = 1$$

for $z \in]0, 1[$ (and $f(z) = 0$ otherwise). Hence, Z is uniformly distributed on the interval $[0, 1[$.

(c) For $x \mapsto cx^{-4}$ ($x \in [1, \infty[$) to be a density, it must satisfy $\int_1^\infty cx^{-4} dx = 1$. Since

$$\int_1^\infty cx^{-4} dx = -\frac{c}{3} x^{-3} \Big|_1^\infty = \frac{c}{3},$$

$\boxed{c=3}$ is the only value of c that guarantees $\int_1^\infty cx^{-4} dx = 1$. We also check that $cx^{-4} = 3x^{-4} \geq 0$ for every $x \in [1, \infty[$. Hence, $c = 3$ is the only value of c that makes $x \mapsto cx^{-4}$ ($x \in [1, \infty[$) a density.

(d) We have

$$\begin{aligned}\mathbb{E}(X) &= \int_1^\infty x \cdot 3x^{-4} dx = \int_1^\infty 3x^{-3} dx = -\frac{3}{2}x^{-2} \Big|_1^\infty = \boxed{\frac{3}{2}}, \\ \mathbb{E}(X^2) &= \int_1^\infty x^2 \cdot 3x^{-4} dx = \int_1^\infty 3x^{-2} dx = -3x^{-1} \Big|_1^\infty = \boxed{3}, \\ \text{Var}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = 3 - \frac{9}{4} = \boxed{\frac{3}{4}}, \\ \mathbb{E}(X^3) &= \int_1^\infty x^3 \cdot 3x^{-4} dx = \int_1^\infty 3x^{-1} dx = 3 \log(x) \Big|_1^\infty = \boxed{\infty}.\end{aligned}$$

(e) To compute the density of $Z = 1 - X^{-3}$, first note that

$$\mathbb{P}\{X \leq x\} = \int_1^x 3y^{-4} dy = -y^{-3} \Big|_1^x = 1 - x^{-3}$$

for $x \geq 1$. Hence,

$$\begin{aligned}\mathbb{P}\{Z \leq z\} &= \mathbb{P}\{1 - X^{-3} \leq z\} = \mathbb{P}\{1 - z \leq X^{-3}\} \\ &= \mathbb{P}\{X \leq (1 - z)^{-\frac{1}{3}}\} = 1 - (1 - z)^{(-\frac{1}{3})(-3)} = z\end{aligned}$$

for $z \in [0, 1]$. (Note that $z \in [0, 1[$ if and only if $(1 - z)^{-\frac{1}{3}} \in [1, \infty[$.) Hence, as in part b., the density f of Z is given by

$$\boxed{f(z) = \frac{d}{dz} \mathbb{P}\{Z \leq z\} = 1}$$

for $z \in]0, 1[$ (and $\boxed{f(z) = 0}$ otherwise). Hence, Z is uniformly distributed on the interval $]0, 1[$.

(f) Note that $Z = F(X)$ is a random variable taking values in $]0, 1[$. Since F is a strictly increasing function, its inverse F^{-1} is a well-defined function, that is,

$$z = F(x) \Leftrightarrow x = F^{-1}(z).$$

Hence,

$$\mathbb{P}\{Z \leq z\} = \mathbb{P}\{F(X) \leq z\} = \mathbb{P}\{X \leq F^{-1}(z)\} = F(F^{-1}(z)) = \boxed{z}$$

for $z \in]0, 1[$. Hence, as in parts (b) and (e), Z is uniformly distributed on the interval $]0, 1[$. Note that the cumulative distribution functions of X in parts (b) and (e) are also strictly increasing (on $[0, 1]$). Indeed, we have $F(x) = 1 - e^{-\lambda x}$ in part (b) and $F(x) = 1 - x^{-3}$ in part (e). Hence, $Z = F(X)$ in both parts. Therefore, the random variables in parts (b) and (e) can be seen as the special cases of the result here.

Q. 2 (Sibling rivalry). a. We have

$$\begin{aligned}\mathbb{P}(X_1 = X_2) &= \int_{\text{over } x} \mathbb{P}(X_1 = X_2 | X_2 = x) \mathbb{P}(X_2 \in dx) \\ &= \int_{\text{over } x} \mathbb{P}(X_1 = x) \mathbb{P}(X_2 \in dx),\end{aligned}$$

the first equality following from the tower property and the second equality from independence of X_1 and X_2 . As X_1 is a continuous random variable, we have $\mathbb{P}(X_1 = x) = 0$ for any x . Therefore, $\mathbb{P}(X_1 = X_2) = 0$.

b. We want to compute $\mathbb{P}(X_1 > 2X_2)$. Note that computing $\mathbb{P}(X_1 \geq 2X_2)$ would give the same result, which is expected as it can be shown that $\mathbb{P}(X_1 = 2X_2) = 0$ with the same argument as in a. Let $\lambda = (5.5)^{-1} = \frac{2}{11}$. Using independence of X_1 and X_2 in the second equality and the density of exponential random variables in the next ones, we get

$$\begin{aligned}\mathbb{P}(X_1 > 2X_2) &= \int 1_{x_1 > 2x_2} \mathbb{P}(X_1 \in dx_1, X_2 \in dx_2) \\ &= \int 1_{x_1 > 2x_2} \mathbb{P}(X_1 \in dx_1) \mathbb{P}(X_2 \in dx_2) \\ &= \int_0^\infty \int_{2x_2}^\infty \lambda e^{-\lambda x_1} \lambda e^{-\lambda x_2} dx_1 dx_2 \\ &= \int_0^\infty e^{-2\lambda x_2} \lambda e^{-\lambda x_2} dx_2 \\ &= \int_0^\infty \lambda e^{-3\lambda x_2} dx_2 \\ &= \frac{1}{3}.\end{aligned}$$

c. Here we want to compute $\mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2))$. We propose two solutions.

Using the law of total probability, $\mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2))$ equals

$$\begin{aligned}&\mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2), X_1 > X_2) \\ &\quad + \mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2), X_2 > X_1) + 0 \\ &= \mathbb{P}(X_1 > 2X_2, X_1 > X_2) + \mathbb{P}(X_2 > 2X_1, X_2 > X_1),\end{aligned}$$

the 0 summand following from $\mathbb{P}(X_1 = X_2) = 0$.

We have $\{X_1 > 2X_2\} \subseteq \{X_1 > X_2\}$ as X_2 is positive almost surely (it follows an exponential distribution), and similarly $\{X_2 > 2X_1\} \subseteq \{X_2 > X_1\}$. So $\mathbb{P}(X_1 > 2X_2, X_1 > X_2) = \mathbb{P}(X_1 > 2X_2)$ and $\mathbb{P}(X_2 > 2X_1, X_2 > X_1) = \mathbb{P}(X_2 > 2X_1)$.

X_1 and X_2 are i.i.d., hence the couples (X_1, X_2) and (X_2, X_1) have the same distribution. Consequently, $\mathbb{P}(X_1 > 2X_2) = \mathbb{P}(X_2 > 2X_1)$. Then

$$\begin{aligned} & \mathbb{P}(X_1 > 2X_2, X_1 > X_2) + \mathbb{P}(X_2 > 2X_1, X_2 > X_1) \\ &= \mathbb{P}(X_1 > 2X_2) + \mathbb{P}(X_2 > 2X_1) \\ &= 2\mathbb{P}(X_1 > 2X_2) \\ &= \frac{2}{3}, \end{aligned}$$

where the last equality follows from b. So

$$\mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2)) = \frac{2}{3}.$$

Alternative solution: The event $\{\max(X_1, X_2) > 2 \min(X_1, X_2)\}$ is equal to the union $\{X_1 > 2X_2\} \cup \{X_2 > 2X_1\}$. Moreover, $\{X_1 > 2X_2\} \cap \{X_2 > 2X_1\} = \emptyset$. Therefore

$$\mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2)) = \mathbb{P}(X_1 > 2X_2) + \mathbb{P}(X_2 > 2X_1).$$

X_1 and X_2 are i.i.d., hence the couples (X_1, X_2) and (X_2, X_1) have the same distribution. Consequently, $\mathbb{P}(X_1 > 2X_2) = \mathbb{P}(X_2 > 2X_1)$, and

$$\mathbb{P}(\max(X_1, X_2) > 2 \min(X_1, X_2)) = \frac{2}{3}.$$

Q. 3 (Men and women). Let us denote by X the height of a random person in the US and denote by Y the gender of this person (m for man, w for woman). The modelling assumptions can be written formally as follows:

$$\mathbb{P}\{X \in dx | Y = m\} = \lambda_1 e^{-\lambda_1 x} dx, \quad \mathbb{P}\{X \in dx | Y = w\} = \lambda_2 e^{-\lambda_2 x} dx,$$

for $x \geq 0$, where $\lambda_1 = (5\frac{3}{4})^{-1} = \frac{4}{23}$, $\lambda_2 = (5\frac{1}{3})^{-1} = \frac{3}{16}$. Besides,

$$\mathbb{P}\{Y = w\} = p = 0.51, \quad \mathbb{P}\{Y = m\} = 1 - p = 0.49.$$

a. The average height of a person in the US is

$$\begin{aligned} \mathbb{E}(X) &= \mathbb{E}(X | Y = m) \mathbb{P}\{Y = m\} + \mathbb{E}(X | Y = w) \mathbb{P}\{Y = w\} \\ &= \frac{23}{4} \cdot (0.49) + \frac{16}{3} \cdot (0.51) \\ &\approx 5.5375. \end{aligned}$$

b. We have

$$\begin{aligned} \mathbb{P}\{X \in dx\} &= \mathbb{P}\{X \in dx | Y = m\} \mathbb{P}\{Y = m\} + \mathbb{P}\{X \in dx | Y = w\} \mathbb{P}\{Y = w\} \\ &= \lambda_1 e^{-\lambda_1 x} dx \cdot (1 - p) + \lambda_2 e^{-\lambda_2 x} dx \cdot p \\ &= ((1 - p) \lambda_1 e^{-\lambda_1 x} + p \lambda_2 e^{-\lambda_2 x}) dx \\ &= \left((0.49) \frac{4}{23} e^{-\frac{4x}{23}} + (0.51) \frac{3}{16} e^{-\frac{3x}{16}} \right) dx, \end{aligned}$$

for $x \geq 0$.

c. Let $x \geq 0$. Recalling conditioning on a continuous random variable, we have

$$\begin{aligned}\mathbb{P}\{Y = w|X = x\} &= \frac{\mathbb{P}\{X \in dx, Y = w\}}{\mathbb{P}\{X \in dx\}} \\ &= \frac{\mathbb{P}\{X \in dx|Y = w\}\mathbb{P}\{Y = w\}}{\mathbb{P}\{X \in dx\}} \\ &= \frac{p\lambda_2 e^{-\lambda_2 x} dx}{((1-p)\lambda_1 e^{-\lambda_1 x} + p\lambda_2 e^{-\lambda_2 x}) dx} \\ &= \frac{p\lambda_2 e^{-\lambda_2 x}}{(1-p)\lambda_1 e^{-\lambda_1 x} + p\lambda_2 e^{-\lambda_2 x}}.\end{aligned}$$

Plugging in $x = 5\frac{1}{2} = \frac{11}{2}$ yields

$$\mathbb{P}\{Y = w|X = \frac{11}{2}\} = \frac{(0.51)\frac{3}{16}e^{-\frac{3}{16} \cdot \frac{11}{2}}}{(0.49)\frac{4}{23}e^{-\frac{4}{23} \cdot \frac{11}{2}} + (0.51)\frac{3}{16}e^{-\frac{3}{16} \cdot \frac{11}{2}}} \approx 0.510124.$$

Q. 4 (Conditioning on sums). a. Let Z be the random variable denoting the sum $X + Y$. To compute $\mathbb{P}(Z \in dz)$, we first compute $\mathbb{P}(Z \leq z)$. We will differentiate it later.

$$\begin{aligned}\mathbb{P}(Z \leq z) &= \mathbb{P}(X + Y \leq z) \\ &= \int_{\text{over } x} \mathbb{P}(x + Y \leq z|X = x)\mathbb{P}(X \in dx) \\ &= \int_{\text{over } x} \mathbb{P}(Y \leq z - x|X = x)\mathbb{P}(X \in dx) \\ &= \int_{\text{over } x} \mathbb{P}(Y \leq z - x)\mathbb{P}(X \in dx),\end{aligned}$$

X and Y being independent. We distinguish between two cases: $z \leq 1$ and $z > 1$ (to see why we need to distinguish between the cases, draw a picture of all the bounds and consider the region where they intersect).

- Consider $z \leq 1$. We know that because Y is uniform, $\mathbb{P}(Y \leq z - x) = z - x$ whenever $z - x \in [0, 1]$. But since $z \leq 1$ it is clear that $z - x \leq 1$ for any $x \in [0, 1]$. The other inequality, $z - x \geq 0$ gives us $x \leq z$. Hence,

$$\begin{aligned}\int_{\text{over } x} \mathbb{P}(Y \leq z - x) \mathbb{P}(X \in dx) &= \int_{\text{over } x} \mathbb{P}(Y \leq z - x) dx \\ &= \int_0^z (z - x) dx \\ &= \frac{z^2}{2}\end{aligned}$$

- Now let's consider $z > 1$. Again, $\mathbb{P}(Y \leq z - x) = z - x$ whenever $z - x \in [0, 1]$. It's clear that $z - x \geq 0$ whenever $x \in [0, 1]$, but now we have to worry about when $z - x \geq 1$, in which case $\mathbb{P}(Y \leq z - x) = 1$. This happens whenever $x \leq z - 1$. When $x \geq z - 1$, we have the usual $z - x$.

$$\begin{aligned} \int_{\text{over } x} \mathbb{P}(Y \leq z - x) \mathbb{P}(X \in dx) &= \int_0^1 \mathbb{P}(Y \leq z - x) dx \\ &= \int_0^{z-1} 1 dx + \int_{z-1}^1 (z - x) dx \\ &= -\frac{z^2}{2} + 2z - 1. \end{aligned}$$

Putting it together, we have

$$\mathbb{P}(Z \leq z) = \begin{cases} 0 & \text{if } z < 0 \\ \frac{z^2}{2} & \text{if } 0 \leq z \leq 1 \\ -\frac{z^2}{2} + 2z - 1 & \text{if } 1 < z < 2 \\ 1 & \text{if } 2 \leq z \end{cases}$$

This gives

$$\mathbb{P}(Z \in dz) = \begin{cases} z dz & \text{if } 0 \leq z \leq 1 \\ (2 - z) dz & \text{if } 1 < z < 2 \\ 0 & \text{otherwise} \end{cases}$$

- b. First let us present a solution that uses the result of question a., and then we will proceed to show a much simpler short solution. Notice that the events $\{Z = 2\} = \{X = 1, Y = 1\}$ and $\{Z = 0\} = \{X = 0, Y = 0\}$, so the conditional expectation $\mathbb{E}[X|Z = z]$ is 1 and 0 for these two cases, respectively. Now, let us consider when $z \in (0, 2)$.

$$\mathbb{E}[X|Z = z] = \int_{\text{over } x} x \mathbb{P}(X \in dx|Z = z)$$

But

$$\begin{aligned} \mathbb{P}(X \in dx|Z = z) &= \frac{\mathbb{P}(X \in dx, Z \in dz)}{\mathbb{P}(Z \in dz)} \\ &= \frac{\mathbb{P}(Z \in dz|X = x) \mathbb{P}(X \in dx)}{\mathbb{P}(Z \in dz)} \end{aligned}$$

We have that

$$\begin{aligned} \mathbb{P}(X \in dx) &= \begin{cases} 1 dx & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \\ &= \mathbf{1}_{x \in [0, 1]} dx \end{aligned}$$

$$\begin{aligned}\mathbb{P}(Z \in dz|X = x) &= \begin{cases} 1dz & \text{if } x \leq z \leq 1+x \\ 0 & \text{otherwise} \end{cases} \\ &= \mathbf{1}_{z \in [x, x+1]} dz\end{aligned}$$

Using the density of Z derived in question a., this gives

$$\mathbb{P}(X \in dx|Z = z) = \begin{cases} \frac{1}{2-z} \mathbf{1}_{\{x \geq z-1\}} dx & \text{if } 0 \leq x \leq 1 \text{ and } 1 < z < 2 \\ \frac{1}{z} \mathbf{1}_{\{x \leq z\}} dx & \text{if } 0 \leq x \leq 1 \text{ and } 0 < z \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Now

- $0 < z \leq 1$:

$$\begin{aligned}\mathbb{E}[X|Z = z] &= \int_0^1 x \mathbb{P}(X \in dx|Z = z) \\ &= \frac{1}{z} \int_0^1 x \mathbf{1}_{x \leq z} dx \\ &= \frac{1}{z} \int_0^z x dx \\ &= \frac{z}{2}\end{aligned}$$

- $1 < z < 2$:

$$\begin{aligned}\mathbb{E}[X|Z = z] &= \int_0^1 x \mathbb{P}(X \in dx|Z = z) \\ &= \frac{1}{2-z} \int_0^1 x \mathbf{1}_{x \geq 1-z} dx \\ &= \frac{1}{2-z} \int_{1-z}^1 x dx \\ &= \frac{1}{2-z} \left(\frac{1}{2} - \frac{(1-z)^2}{2} \right) \\ &= \frac{1}{2-z} \left(z - \frac{z^2}{2} \right) \\ &= \frac{z}{2}\end{aligned}$$

$\mathbb{E}[X|Z = z] = \frac{z}{2}$

Alternative solution: we can solve this by the following observation. Note that since X, Y are i.i.d., the pairs $(X, X + Y)$ and $(Y, X + Y)$ have the same distribution. Thus, the random variables $X|X + Y = z$ and $Y|X + Y = z$ have the same distribution as well and therefore we have that $\mathbb{E}[X|X + Y = z] = \mathbb{E}[Y|X + Y = z]$. On the other hand note that:

$$\mathbb{E}[X|X + Y = z] + \mathbb{E}[Y|X + Y = z] = \mathbb{E}[X + Y|X + Y = z] = z.$$

Consequently,

$$\mathbb{E}[X|X + Y = z] = \mathbb{E}[Y|X + Y = z] = \frac{z}{2}.$$

Q. 5 (Telemarketers). We assume that the number of customers N is Poisson distributed with mean 30. Denote the length of the i th call by X_i for each $i \in \{1, 2, \dots\}$. Hence, the amount of time S it takes for the telemarketer to complete all calls is given as the random sum

$$S = X_1 + \dots + X_N = \sum_{i=1}^N X_i.$$

For $n > 0$, conditional on $N = n$, the length of each call will be exponentially distributed with parameter n . That is,

$$\mathbb{P}\{X_i \in dx | N = n\} = ne^{-nx}dx, \quad x \geq 0.$$

In particular,

$$\mathbb{E}(X_i | N = n) = \frac{1}{n}.$$

Therefore, given $N = n$, the expected amount of time it takes to complete all calls is

$$\begin{aligned} \mathbb{E}(S | N = n) &= \mathbb{E}\left(\sum_{i=1}^N X_i \mid N = n\right) \\ &= \mathbb{E}\left(\sum_{i=1}^n X_i \mid N = n\right) \\ &= \sum_{i=1}^n \mathbb{E}(X_i | N = n) \\ &= n \times \frac{1}{n} = 1. \end{aligned}$$

Note that $\mathbb{E}(S | N = 0) = 0$ and $\mathbb{E}(S | N = n) = 1$ for all $n > 0$. Then, the expected amount of time it takes for the telemarketer to complete all calls for the day is

$$\mathbb{E}(S) = \sum_{n=0}^{\infty} \mathbb{E}(S | N = n) \mathbb{P}\{N = n\} = \sum_{n=1}^{\infty} 1 \cdot \mathbb{P}\{N = n\} = 1 - \mathbb{P}\{N = 0\} = 1 - e^{-30}.$$